

Absolutism, Relativism, and Universalism in Personality Traits Across Cultures: The Case of the Big Five

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Abstract

Personality is a broad concept used to organize the myriad ways that people differ psychologically from one another. There is evidence that such differences have been important to humans everywhere, in that personality-relevant terms appear in all known languages. Empirical attempts to identify the most useful individual differences and their structure have emphasized cross-cultural evidence, but rigid adherence to a Big Five model has sometimes meant ignoring heterogeneous results. We start with a framework for more precisely defining the universality versus cultural-specificity of personality concepts and models in order to better assess cross-cultural evidence. As this 50th anniversary of the *IACCP* is also the 50th anniversary of the first large lexical study of personality and more or less of the Big Five model, we take the opportunity to explore both how personality has been studied across contexts using the lexical method, and in 100 articles on personality topics (most using questionnaires) that were identified in the pages of *JCCP*. Personality articles in *JCCP*, classified into three types based on their balance of emic and etic components, illustrate larger trends in personality psychology. With the benefit of hindsight, we reflect on what each type has to offer going forward, and we encourage cross-cultural personality psychologists to go beyond imposed etic studies that seek primarily to confirm Western models in other contexts. The kinds of insights that more integrative emic and etic approaches can bring to the study of psychology across cultures are highlighted, and a future research agenda is provided.

Keywords

Big Five, lexical studies of personality, psycholexical studies, cultural psychology

Personality is a broad concept used to organize the many ways that people differ psychologically from each other. It can be defined as an individual's characteristic patterns of thinking, feeling, motivation, and behavior, together with the mechanisms directly underlying those patterns

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(Funder, 2001). There is evidence that making distinctions about such differences has been important to humans everywhere for a long time, in that character- and personality-relevant terms appear in all known languages (e.g., Dixon, 1982). What is less clear is the extent to which the same or different distinctions are useful and important across contexts.

It is crucial to remember, in the study of personality, that human beings, by definition, exist in a cultural context. Unique among animals, we have evolved to be dependent on cumulative social learning for our physical survival (Henrich, 2016). This reality permeates every aspect of our lived experience. In addition to artifacts and practical knowledge, culture shapes our brains, biology, and psychology (Henrich, 2020), including the values, standards, norms, and motivations that are shared among individuals in a society (e.g., a nation or a language-community), and transmitted across time and generations, although what is more or less shared is subject to continual change (Saucier, in press). Thus, culture and psychology are substantially interwoven. A focus on culture means looking at shared psychological patterns that differ between groups. A focus on personality instead emphasizes differences between individuals, which emerge and are experienced within a cultural matrix, but can include differences in how cultural ideals are internalized or endorsed (Leung & Cohen, 2011). Traits are differentially permitted, celebrated, noticed and discussed across contexts, and they interact with local values, reward systems, and possible roles, leading to different outcomes.

While studying personality and culture together has its complexities and challenges, we believe that the interwoven nature of the two domains means that doing so is necessary: The study of personality is problematically incomplete without consideration of the context. In the review that follows, we look back at the ways that personality has been studied across cultures, through the lens of two 50-year anniversaries: that of the *International Association for Cross-Cultural Psychology (IACCP)* and that of significant lexical studies of personality and (more or less) that of the Big Five model. In all cases we highlight the tension between attempts to take cultural differences into full consideration, and an expedient tendency to over-impose Western models, which reflects the dominant trend of this period in psychology (Henrich et al., 2010; Thalmayer, Job et al., 2020). After providing a framework for defining degrees of universality versus cultural specificity in personality, we review the history of lexical studies of personality and how they shaped the Big Five model that has dominated this field. We then consider the types of personality research (often using personality questionnaires) that has appeared in the pages of *Journal of Cross-Cultural Psychology (JCCP)*. With the benefit of hindsight, we reflect on what different types of studies have taught us in the last 50 years, and what they could teach us going forward into the next 50 years, considering important questions that remain to be addressed.

Local Versus Imported Constructs: Degrees of Specificity versus Universality

Human beings are very adaptable. Our species spread to every corner of the globe, and developed cultural systems that help us thrive in a wide variety of environments (Henrich, 2016). In turn, adaptability now allows for globalization, as certain languages and cultural norms spread to facilitate interconnection. Thus, human adaptability can be a double-edged sword, leading to both the development and the loss of rich cultural and linguistic diversity. In personality psychology we encounter these same forces. Methods exist, in particular lexical studies, which can be used to identify the traits of most importance to a specific context, and to compare them to those identified in other contexts quantitatively. This creates potential to understand both culture-specific aspects of personality description and structure and to define universal aspects. Too often, however, traits and structural models of interest in the West are simply exported for use in other contexts.

We therefore want to begin our review by emphasizing the key difference between discovering a trait in a population (through indigenous lexical studies or other bottom-up approaches) versus importing a trait into a new context, which is often done through questionnaire research). While it is those traits that have been important to a population that are likely to be encoded into a single word in the local language (Goldberg, 1981), it is generally possible to take the meaning of a trait from one place to another. For example, in Japanese the term *amae* describes a pattern of desire, emotion, and behavior related to being indulged by someone else, which can be positive, as in seeking to be indulged a close relationship, or negative, if one is demanding too much (Behrens, 2004). Once this meaning has been explained, people outside Japan may be able to think of people who behave this way, and may enjoy having a new concept for describing human variation. But knowing that this term comes from Japan also tells us something interesting about Japanese culture: a glimpse into how people may relate to each other, and what they notice and say about each other's relations. We discuss below the ways that Extraversion may be analogous to this, as a trait domain that is especially valued in and perhaps descriptive of society in the United States, and we note distinctive traits that have emerged in recent lexical studies in African languages.

When we export personality traits, what do we gain and what do we lose? Assessing people on the same constructs across contexts can allow for comparison and for integrating research results. And a personality model created in one place can often be used in others: people are similar enough that we can understand each other's distinctions once they are explained. But the traits that are the most frequently used within a population, and that have been encoded into the language, are the differences that have been the most noticeable and useful locally. This means that they don't only tell us about personality differences, but also about the local society. We lose something interesting and meaningful in a rush toward the expediency of homogenization.

It is perhaps useful here to make distinctions in terms of the degree and extent to which a concept is universal versus culturally specific. In 1994 Adamopoulos and Lonner sought to define absolutism, relativism, and universalism in the study of psychology across cultures. Heine (2020); based generally on Norenzayan & Heine, 2005) distinguishes four levels of universality, which are applicable to the personality domain to some extent. He starts with *non-universals*, which refer to culture-specific phenomena that are not really translatable. In the trait domain this could refer to characterizations so specific to a context that they cannot be translated, but this category is likely to be rare for personality concepts, as these usually focus on general qualities and patterns.

In the larger category of "weak" universals are traits and concepts that can be imported through translation and explanation, though they are not encoded into a single commonly used term in most languages. These concepts can vary in their relevance, associated meanings, and accessibility. Heine breaks this level of universality (psychological phenomena that exist to some minimal extent everywhere) into two categories, based on whether they are used for different or the same purposes. Fontaine (2011) better adapts this distinction to the domain of personality, defining the same function as present when a trait or process has the same underlying meaning and an identical nomological network across contexts. Thus, we can say that a personality concept (either a single-word concept or a trait construct represented by multiple descriptors) is an *existential universal* if it is universally translatable, but not always with the same meaning and associations. We can say it is a *functional universal* if it has very similar meaning and associations across cultures.

An illustrative example of a weak universal might be "face." This East Asian concept can be reasonably translated into virtually any language, but it lacks the same meaning and associations in other contexts. It could thus be considered an existential but not a functional universal. If Extraversion is shown to be universally translatable and moreover to have similar meaning and associations everywhere—including independence from other broad personality traits as in the

Big Five model—it could qualify as a functional universal, even if was not equally accessible or salient in all cultural contexts. But if its evaluative valence and associations differ across contexts, for example as has been seen for its association with well-being (Kim et al., 2018), it would likewise be an existential but not a functional universal.

Accessibility universals are strong cross-cultural universals: They are not only used in similar ways across cultures but equally accessible in all of them, and because of these properties they (alone) can be used to compare cultures and populations. Fontaine (2011) makes a distinction between accessibility universals that arise due to cultural or environmental causes (repertoire universals) and those that arise due to biological or genetic causes (absolutist universals). Regardless of that distinction, stringent thresholds must be met to be considered an accessibility universal; Fontaine regards measurement equivalence as a requisite. Status as an accessibility universal means that the same instrument can justifiably be used to measure the psychological trait(s) or process(es) across cultures. Allik and Realo (2017) argue that few truly universal traits can be found, and more often we must settle for “near-universals”; if the exceptions are rare, this may indeed be enough to be counted as an accessibility universal.

Personality concepts identified among single terms across 12 diverse languages (a set of languages unlikely to have influenced each other), including “disobedient,” “gossips,” and “foolish” (Saucier, Thalmayer, & Bel-Bahar, 2014), are candidates for universal concepts, if they can be shown to have similar evaluative and predictive associations. At the trait dimension level, the Big Two dimensions of Social-Self-Regulation and Dynamism, which emerge spontaneously in the content of the first two factors of lexical studies across diverse languages (Saucier, Thalmayer, & Bel-Bahar, 2014) or the Pan-Cultural Three dimensions of Affiliation, Order, and Dynamism, which appeared as the core of 11 taxonomies (De Raad et al., 2010) are candidates to form a universally accessible structural model of personality, if measurement models with appropriate levels of measurement invariance can be established.

There is less basis to consider the Big Five as an accessibility universal: First because it has a more checkered record of confirmation by the criterion of appearing spontaneously in the lexicons of diverse languages, a kind of universal salience that can be regarded as a minimal sort of universal equal accessibility, akin to configural invariance. Secondly, in studies using Big Five questionnaires, the model lacks appropriate measurement equivalence (Laajaj et al., 2019).

Costa and McCrae, in a 1997 *American Psychologist* article, advanced the claim that the five factors are not only equally accessible everywhere, but that this is because they are genetically determined endogenous traits. This injects, in Fontaine’s terms, absolutism into the claim of universality. The assumption is then made that the same instrument can be used across cultural contexts to capture these absolute trait universals. Their NEO inventory does appear to translate reasonably well, despite its reference to culture-specific contexts such as museums and Las Vegas, but as we touch on later, limits to the reliability and reproducibility of the structure and its factors, not to mention the lack of strong evidence for measurement invariance, cast doubt on claims of strong universality for the Big Five. It would be safer to say that this model appears to involve existential universals that may also have some claim as functional universals.

Comparisons of the results of emic studies can best establish how universally salient a trait or process is. As Church and Lonner pointed out in a 1998 *JCCP* review of cross-cultural personality research, “imposed-etic” studies may optimize the chances of finding cross cultural comparability while excluding culturally specific dimensions. Thus, the most persuasive evidence comes from indigenous (emic) studies, for example from lexical studies.

Lexical Studies of Personality and the Big Five

Lexical studies of personality are based on a rationale from linguistics: Where an idea is expressed often, it will become encoded into a single word. Sedimented into the language are the individual

differences that are most useful and salient to the community of speakers. After the idea was proposed as way to identify an objective model or personality trait structure, initial catalogs were made in English and in German (Allport & Odbert, 1936; Baumgarten, 1933). Goldberg (1981), the first to apply modern computing capacity to large scale tests of the idea, early emphasized the cultural corollary of this idea: the more important an individual difference is for humans in general, the more languages should have a word for it. He laid out a research program for personality psychology, with an emphasis on lexical studies and on the identification of universal aspects of structure (1981).

Interestingly, 1972, the year of *IACCP's* founding, was also the year Goldberg collected the first large dataset of 1,710 person-descriptive adjectives in English, which was not only the first of its type but still counts as one of the largest lexical sets administered. It was completed by about 200 students in the United States, and, thanks to Warren Norman, to 123 in Australia (combined in data analyzed by Ashton et al. [2004]).

Lexical studies of personality are conducted in four basic steps, which are transferable to any language with a lexicon compiled in writing. First, all terms used to describe psychological differences are extracted from a comprehensive lexicon; the list is then reduced to a tractable number by removing redundant and less familiar terms; the list is administered to participants, asking them how well each term describes a target (the self or a well-known other); factor analysis is used to determine which terms group together into traits, and which best distinguish between individuals in the population (Thalmayer et al., 2021). This methodology has the potential to be culturally neutral, and to elegantly and systematically combine emic (indigenous, bottom-up) and etic (imported, allowing for specific cross-cultural comparisons) elements, as recommended by leaders in cross-cultural psychology (e.g., Cheung et al., 2011). Using the terms embedded in the lexicons of different languages allows for building an objective local model without imposing or ignoring anything. In a separate step, translated terms can be included where needed to create marker scales for etic models, which can be correlated with emic factors for empirical tests of replication (e.g., Thalmayer, Job et al., 2020). There has not been a consensual standard for a threshold of cultural specificity, but an emic factor for which less than 25% of the variance can be explained by a set of imported etic factors might be considered. Of the potentially beyond-the-Big-Five constructs studied by Saucier and Goldberg (1998), many of those with less than 25% variance accounted for by the Big Five appeared as independent factors in structures that allowed for high-dimensionality (Saucier & Iurino, 2020), which suggests that a useful criterion for independence of factors may lie near this threshold.

It would not be quite true, however, to say that the potential for elegant and systematic integration of emic and etic elements has always characterized the history of published lexical studies. They have sometimes been less exploratory and more confirmatory in emphasis than they might have been. A five-factor structure identified by Norman in the 1960s was already being referred to as the “Big Five” by Goldberg (1981); in this influential chapter on language and individual differences, after a long review of the possibilities of learning about structure via language, Goldberg makes a firm landing on five factors based on what some studies in the English language (alone) had found. A Big Five model was also already being explored by McCrae and Costa (1992), leading eventually to the NEO-PI-R (1992) Goldberg's first widely-available publication using the 1972 data appeared in 1990 after a long process of creating clusters, necessitated by computational limitations that limited the number of variables that could be analyzed together. This meant that he analyzed 75 clusters rather than 1,710 variables, and later evidence suggests that this sharp variable-reduction played a major role in the identification of five factors, rather than more (Saucier, in press; Saucier & Iurino, 2020). Varimax rotation tends to constrain the number of factors, while ipsatized data leads to more bipolar factors, which are more prone to resemble desired five or six factor structures; in contrast, other methodological variations

(rotation type and data treatment) demonstrably lead to very different structural solutions in Goldberg's data (Saucier & Iurino, 2020).

Goldberg's (1981, 1990) papers set the conceptual grounds for a series of lexical studies in many languages. Studies were soon completed in Dutch (De Raad et al., 1992) and German (Angleitner et al., 1990), which made some helpful modifications to the methodology, for example introducing taxonomic schemes to reduce subjectivity in the selection, reduction, and classification of terms, that influenced later studies. While these studies more or less confirmed the Big Five structure, there was a lack of replication of the factor variously labeled as Intellect, Openness, or something else. These studies were followed by efforts in 20 more languages by 2018 (15 in Indo-European languages, all but three recent African-language studies using college students; see Thalmayer, Job et al., 2020 Supplemental materials Table S1 for a list). Until three recent studies, virtually all were conducted using the same analytic approach, including varimax rotation on ipsatized data, which, intentionally or not on the part of the researchers, shaped the number of factors identified to be five or six (the impacts of these choices are discussed and demonstrated in detail in Saucier & Iurino, 2020; Thalmayer, Job et al., 2020). In more than half of the studies, only five factors were given any serious attention.

The choice of factor-number has been traditionally justified based on applying a visual scree test to a graph of unrotated-factor eigenvalues, and seeing a noteworthy drop in eigenvalue size (a small cliff in the scree) after five or six factors. The scree test operates on the implicit assumption that the data has primarily a few strong common factors, and conclusions from it are somewhat subjective (Kaiser, 1960); which differences in eigenvalues appear noteworthy can depend on how the graph is scaled or whether it is oriented horizontally or vertically. There has been a lack of attention to alternatives, such as parallel analysis (Horn, 1967). Another rule for determining the number that has been mentioned in the literature is that enough factors should be extracted to account for 60% of the variance in the correlation matrix (Hair et al., 2017), while five- and six-factor solutions in lexical studies have rarely accounted for more than 25%.

Once results were obtained in these studies, correspondence to the Big Five was in several studies (6 of the 27 in the table cited above, most of which were published in the 1990s when there were, of course, fewer options for quantitative comparison) only based on visual inspection of similarity of content. Emic exploration, which would mean bringing more interest to local variations in frequently used terms and in the dimensions beyond the first few that arise naturally in the data, was minimal; generally only models close to five were extracted and discussed. Thus, while having data from many contexts has created a sense of cross-cultural replication for the Big Five, on closer inspection we can see that norms and traditional standards in the field have meant that we have not really learned as much from this diverse data as we might have, about the personality terms of most interest in the various places it was collected.

In a scientific zeitgeist where settling on a consensus model was highly desirable, the Big Five (Extraversion, Agreeableness, Conscientiousness, Emotional Stability/Neuroticism, Intellect/Openness) appeared to be supported in several (though not all) of the lexical studies that followed a strict method-template. This useful model has many merits, for example, good predictive validity in the West, though the effect-sizes are not typically large. But early methodological choices (clustering, ipsatized data, varimax rotation, scree plot inspection, accepting less than 50% variance explained) have tended to be imitated without much reflection, and these have constrained the number of factors and helped guide results toward replicating a five-factor model.

Lexical work published in *JCCP* illustrates the tensions that arise when unique and divergent structures emerging from emic-oriented analyses intersect with the motivations (inadvertent or not) of investigators to confirm a popular etic model. Wasti et al. (2008) investigated personality dimensions in the Turkish language. Before re-analyzing earlier data using 498 Turkish adjectives from Goldberg and Somer (2000), the team reduced the set to 434 adjectives, representing stricter criteria for the personality domain, another important criterion for finding the Big Five

(e.g., Saucier, 1997; Saucier, Thalmayer, Payne, et al., 2014). In fact, Goldberg and Somer (2000) had done something analogous, after the original set of 498 adjectives failed to confirm the Big Five. These strategies serve to make *a priori* dimensions look more inevitable (and thus universal) than the empirical results initially suggest. Similarly, Lee and Ashton's (2009) reanalysis of Greek personality adjectives involved omitting 40 of the 400 original terms that they deemed inadmissible as personality descriptors, thereby arriving at a structure that more closely resembled their intended template model, even more so after an additional step of hand-rotating two of the dimensions to fit pre-existing specifications.

These strategies have their justifications, and we critique methods here not to censure colleagues who certainly approached their work with integrity and dedication to the scientific method. Our goal is to challenge and inspire the next generation of cross-cultural psychologists. We believe that studies of personality across culture can benefit from increased reflexivity about implicit assumptions (i.e., degree of investment in an etic model) and methodological choices. In our view, a stance of explicit curiosity about both more universal and more culturally specific elements of personality may be the best starting place, as welcoming both kinds of findings will make them more likely to be identified in the data. Other open-science initiatives such as pre-registered analysis plans with explicit decision rules for the replication of models are also important. In discussion of the future of research on personality across cultures, below, we consider the kinds of insights that studies following these principles can yield.

Fifty Years of Personality Research in *JCCP*: Too Etic, Too Emic, or Just Right?

While the lexical work described above mostly appeared in other journals, we identified at least 100 articles on the topic of personality (using that term in their titles or keywords) published in the *Journal of Cross-Cultural Psychology* journal during the last 50 years. Of these, only five appeared in the 1970s, and eight in the 1980s. As noted by Church and Lonner (1998), historically the relation of culture and personality was addressed more in anthropology than in psychology. Unsurprisingly then, the first *JCCP* article we found concerning personality used a model associated with mid-20th-century cultural anthropology: the "modal personality" approach to culture-personality relations, explored in this case by Lefley (1972) using a questionnaire and sentence-completion test administered in the Bahamas. The modal personality approach was fading from influence by 1972, and this approach did not become influential in *JCCP* or in personality journals. Allik et al. (2011), in a study of whether there is some modal "Russian soul," provided a quantitatively-based debunking of the essential modal-personality idea, that cultural contexts mold a single convergent personality type.

The pace of personality content in *JCCP* quickened in the 1990s, with nine articles by 1997, and a special issue on "Personality Measurement in Cross-Cultural Perspective" in 1998, which included another 11. About a quarter of articles in the 1990s focused on the Big Five model. In line with larger trends in personality psychology, this proportion greatly increased in the 2000s, as did the total number of articles on personality topics. This illustrates the ways that the Big Five model was generative for the study of personality, by providing more widely-known and accepted concepts. It might also be reasonable to conclude that the identification of a "consensus" model made the topic of personality more interesting to cross-cultural psychologists, as opposed to earlier emic work that may have appealed more to anthropologists.

We found that the articles on personality published in *JCCP* over the last 50 years can be organized with more or less the same distinctions used by Church and Lonner (1998), into three or four broad types based on their balance of emic (indigenous) and etic (imported) components. We describe below some highlights from each of these approaches, we reflect on the worldview indicated by each, and we consider what these approaches have to offer our field going forward (Some interesting related

topics covered in *JCCP*, including response styles and prediction of acculturation, are excluded here due to space limitations). While this portion of our review focuses exclusively on articles published in *JCCP*, we believe these categories and conclusions would be likely to generalize to cross-cultural personality studies published in the *European Journal of Personality* and the Personality Processes and Individual differences subsection of the *Journal of Personality and Social Psychology*.

Imposed Etic Studies: Is It Time to Reduce Their Role?

Conceptually, testing a model from one setting in a new context can be a practical contribution. Importing is cost effective and allows for integration of results. However, imposed-etic studies have come to dominate personality psychology across cultures and rather problematically so. In particular, many studies published in *JCCP* have examined the reproducibility (and thus the generalizability and universality) of Big Five inventories. Some appear to assume the universality of the Big Five model as a given (e.g., MacDonald, 1998) following the bold example of McCrae and Costa PT (1997), who deemed the Big Five structure a human universal on the basis of recovering the NEO's factor structure for translated versions using Procrustes rotation, a finding that does not even imply configural invariance (e.g., Fischer & Karl, 2019). In the pages of *JCCP*, Piedmont and Chae (1997) and later Ispas et al. (2014) assessed the generalizability of NEO inventories in translation to Korean and to Romanian, respectively. The authors similarly conclude that the NEO is generalizable to these contexts, although in both cases some facets had low reliability and the structure deviated from expectations in an unconstrained varimax rotation. The discrepancies were resolved in both studies by turning to a constrained Procrustes rotation approach.

Schmitt et al. (2007) examined the "geographic distribution" of Big Five traits in self-report samples from college students in 56 countries. Reliabilities ranged from .55 to .68 in Africa but from .77 to .84 in North America. To resolve discrepancies between the model and results, the authors created region aggregates and also resorted to Procrustes rotation, in this case not even reporting varimax results. Congruences of Procrustes results, with results from the United States as a reference point for *T*-scores, were lowest for Africa (0.84–0.93), South and Southeast Asia (0.85–0.91) and for the Middle East and Eastern Europe for Agreeableness specifically (0.87–0.88). In other data, convergence of the NEO with the Big Five Inventory was poor for Openness and Agreeableness at the national level, and correlations of Big Five scales with self-esteem varied substantially. The emphasis was on reporting similarities, and little interpretation was brought to these potentially interesting divergences. Furthermore, mean comparisons were made without testing for measurement invariance, which does not meet current methodological standards (e.g., Fischer & Karl, 2019; cf. Welzel et al., 2021), and comparisons did not address reference-group effects. Only when discussing limitations do the authors mention these as a possible reason for Conscientiousness and Emotional Stability being lowest in East Asia, ending with an appropriate caveat: ". . . it seems probable that certain biases and measurement errors of individual assessment devices reduce the accuracy of mean-level portrayals of national personality" (p. 204).

Allik and McCrae (2004) examined patterns of NEO profiles across 36 nations, including 22 from the industrialized West. Contrary to Five Factor Theory (e.g., McCrae & Costa PT, 1997) national-samples' differences, which ran counter to stereotype, were attributed to "features of culture" (p. 13). As in Schmitt et al. (2007), regional differences ran counter to stereotype, likely due to reference-group effects and response biases. Allik et al. (2017) similarly compared mean NEO profiles across 62 countries (including the earlier 36), noting that between-group differences were eight-and-a-half times smaller than within-group differences. Their emphasis on smallness of cross-cultural differences implied support for the universality of the Big Five model, although random data would also yield small differences. They usefully summarize Möttus et al. (2012) work on reference-group effects, noting that counter-intuitive rankings on Conscientiousness have led to lack of trust in national mean scores.

Caprara et al. (2000) explored methods for comparing factor structures across contexts, concluding that EFA and simultaneous component analysis (SCA) generated similar results for a set of four (all Western industrialized) samples, but that CFA indicated more misfit, and may be “oversensitive.” Indeed, CFA fit indices for multi-dimensional inventories with established predictive validity often fail to meet standard SEM benchmarks, likely due to cross-scale covariance, even in their context of origin, suggesting that benchmarks based on other research domains may be unrealistic for personality psychology (Hopwood & Donnellan, 2010). In the era of open science, “reasonable” fit based on that typical for the type of inventory could be established a priori in pre-registered analysis plans, insuring that the claim of oversensitivity does not function as a way for researchers to exclude a method that doesn’t yield desired results. This can also include learning from CFA analyses beyond a dichotomous evaluation of accept or reject, fit or no-fit, for example by planning to discriminate models with modestly good fit from those with terrible fit. Similarly, while measurement invariance analyses allow for confidence in the cross-cultural adaptability of inventories (Fischer & Karl, 2019), other authors have demonstrated limitations to this approach, and advocated instead for alternative CFA approaches coupled with nomological validation (Welzel et al., 2021), which are also useful options.

The use of strategies such as Procrustes rotation in imposed etic research maximize fit with a pre-existing model, which is logical if the goal is to establish universal salience. Unfortunately, however, they inhibit the chance of learning about interesting and relevant discrepancies that relate to cultural differences, or about emic factors that could turn out to be more universal than the one imposed. For example, what if being gossipy is a more universal dimension than Extraversion, or Industriousness more than neatness-oriented Conscientiousness? As noted above, historical conditions led to the need for a shared model and contributed to the dominance of the Big Five. Attempts to establish a universal model also laudably emphasize similarities in the shared human experience. As science evolves, however, we must consider the limitations of our models and our strategies. In hindsight, over reliance on the Big Five is intertwined with a history of erroneously assuming that Western (often white, college educated) samples are normative, and that results can automatically be generalized to the rest of the world (e.g., Henrich et al., 2010; Thalmayer et al., 2021). The evidence from many lines of research is clear: Culture shapes psychology, and people in industrialized societies have been shown to be outliers, unrepresentative of the majority of humans in many regards (Henrich et al., 2010, 2022). By imposing Western etics in cross-cultural research, researchers miss important dimensions of variation, thus leading to an uneven and incomplete understanding of human psychology. Furthermore, the methods of imposed-etic research can problematically assume a reality that is not characteristic for the majority world and, at worst, may be experienced as a kind of scientific colonialism (Adams et al., 2015; Galtung, 1967), reducing the role of majority-world collaborators to data collectors (Saab et al., 2020) on projects that were fully designed before their inclusion.

This kind of research is likely to continue being published in many journals. In our opinion, given its leadership role in cross-cultural psychology and its highly international membership, *JCCP* editors and reviewers could set a higher standard for cross cultural work, and couldn't encourage authors to develop more nuanced assessments of cross-cultural fit, as modeled by the next two types of article.

Nuanced Etic Studies: Learning from the Ways an Imported Model Does and Doesn't Fit

Another kind of etic study has a crucial difference: The authors go beyond a yes-or-no question of whether an inventory works in a new context to more fully assess the ways it does and does not. This can include thoughtful description of differential item functioning, comparison of

methods, and analytic strategies that seek to identify and explore rather than simply to confirm. These studies learn more from the cultural context in question, thus making better use of the data and the contributions of international collaborators.

The first *JCCP* publications to engage with the issue of replicating the structure of an established inventory across contexts were published in the late 1980s. Bijnen et al. (1986; Bijnen & Poortinga, 1988) assessed prior cross-cultural comparisons made with the Eysenck Personality Questionnaire and critiqued their use of a particular (and non-standard) way of computing factor congruence, which likely biased results in a favorable direction.

Huang et al. (1997) investigated the NEO-PI-R in samples of American and Filipino college students, but unlike the studies described above in Romanian and Korean, the authors gave attention to differential item functioning (DIF), with the explicit aim to identify meaningful differences. They reported that about 40% of the items exhibited DIF, though they concluded that only a few would qualify as fully “emic,” meaning a construct with near-zero discriminant validity in outside contexts. In a subsequent step, the authors showed that elimination of these items reduced the number of between-population differences.

Paunonen and Ashton (1998) considered six personality questionnaire platforms that had been translated and used across contexts (not including the Big Five). In addition to the commonly discussed elements of reliability and internal structure, the authors introduced a demanding criterion for cross-cultural generalizability, assessing consistency in the predictive validity of the inventories (for a similar argument, see Bartram, 2013). They concluded that there was a dearth of evidence for this kind of generalizability, which would indeed require a very high degree of consistency across cultures in how outcomes are related to personal dispositions.

Eigenhuis et al. (2015) examined violations of measurement invariance between US and Dutch scores on the Multidimensional Personality Questionnaire (MPQ; Tellegen & Waller, 2008). Whereas Paunonen and Ashton (1998) argued against the importance of strict invariance, emphasizing instead correspondence in factor structure and reliability, these authors carefully examined it. Although the MPQ inventory showed better fit across the two (similar) populations than has been the case with many personality inventories (e.g., those assessed by Hopwood & Donnellan, 2010), a substantial fraction of MPQ items exhibited DIF, potentially attributable to cultural differences.

In our opinion, studies like these meet the same practical need as imposed-etic studies, allowing a useful established inventory to be adapted to a new context. Importantly they also report on divergences that have practical and theoretical implications for understanding personality and its assessment across contexts. Furthermore, they imply a world view that allows for the existence of both human universals in psychology and for the influence of the cultural context in psychological phenomena.

Emic and Etic Components Integrated

We agree with the recommendation made by Cheung et al. (2011) in an influential American Psychologist article, that the combination of emic and etic components is ideal in the study of personality across cultures. This implies systematically seeking both universal and culturally specific aspects. Among the many examples of such work published in *JCCP*, there is great heterogeneity, and some methods may prove more useful than others going forward. In all cases, however, rich description of local aspects of personality traits or structure are combined with useful comparison to familiar, established models.¹

An early example from Church and Katigbak (1988) set out an emic strategy for identifying and assessing personality dimensions, illustrated with details of its application in the Philippines. Under this strategy, emic concepts are identified, incorporated into assessment procedures, then submitted to construct validation studies. Finally, these are related to etic conceptions and measures in order

to “determine whether emerging emic dimensions are unique (emic), comparable (universal, etc), or overlapping (‘partial etic’, etic with emic exemplars)” (p. 152) with personality dimensions from other contexts (Church & Katigbak, 1988). This was an early and unusually clear exposition of the ideal design of research related to lexical studies of personality descriptors, including those authors’ later publications on Philippine personality dimensions (Church et al., 1997). While most lexical-study literature has been published in other journals, it is noteworthy that *JCCP* published this early and helpful specification for such research.

Ten years later, Guanzon-Lapeña et al. (1998) reviewed recent projects involving indigenous Filipino personality constructs and concluded that they were all *potentially* subsumable under Big Five categories. Several of the constructs, however, including thrift, conformity, reflectiveness, and egotism versus humility, were found to be at least somewhat “beyond the Big Five” by Saucier and Goldberg (1998), and appeared as independent dimensions in the analyses of Saucier and Iurino (2020). What is only potentially subsumable under the Big Five may also be independent of it. Here again we note that implicit assumptions based on a popular model may inadvertently influence a sense of correspondence with that model and the visibility of content beyond it, highlighting the importance of establishing *a priori* guidelines.

JCCP also published a key early comparison of structures derived from seven European-language lexical studies in search of a “lingua franca” of personality description (De Raad et al., 1998). Analyses illustrate the distinction between existential/functional universals and the more stringent criteria associated with universal salience and accessibility. When five-factor structures were compared before target rotation, congruence coefficients were mostly of an unacceptable magnitude (only 6 of 30 reached the minimally permissible 0.80 level), averaging 0.68. After target rotation, a procedure that maximizes their correspondence with and minimizes their variation from each other and any common structure, the average increased to 0.76, and 14 of 30 congruences reached 0.80. This early comparison on favorable ground—European languages, including those in which the Big Five was first identified—cast doubt on the universal salience and accessibility of the Big Five, while still allowing for arguments that it is an existential and (depending on whether its meaning and pattern of associations with external variables is consistent) perhaps a functional universal.

At least six *JCCP* articles from the late 1990s through the 2000s explored emic Chinese personality characteristics and reported on the development of the Chinese Personality Inventory (CPAI) as an emic alternative to the five-factor model (FFM; Cheung et al., 1996). Cheung and Leung (1998) described a Chinese Tradition factor (apparently an early label for Interpersonal Relatedness, described below) as independent of the Big Five. An English translation of the CPAI was provided by Cheung et al. (2003), along with evidence that, like the NEO-PI-R, it could be successfully translated and utilized in other contexts. This inventory was examined in two US samples by Lin and Church (2004), who concluded that it was not culture-specific (emic) because it was measurable in an American sample, but that it was more salient among those with a traditional Chinese heritage. Additional emic personality constructs were investigated by Cheung et al. (2001) in terms of whether they fall outside the Big Five. Their focus was on a construct labeled Interpersonal Relatedness, including the interpersonal concerns common among Chinese populations, for example, face, Ren Qing reciprocity, and surface harmony.

A key difference between the FFM and CPAI is Openness, which Cheung and colleagues regard as an etic import. Cheung et al. (2008) concluded that this construct, while translatable to the Chinese context, is not a coherent part of the “Chinese implicit theory and taxonomy of personality” (p. 103). Taken together this work provides examples of two major constructs (Chinese Tradition and Openness) which have culture-specific aspects, but may qualify as existential universals if they are shown to be widely translatable or exportable.

As part of the ambitious South African Personality Inventory (SAPI) project, Valchev et al. (2013) compared implicit personality concepts across a dozen ethno cultural groups, involving

11 languages, in South Africa. They found that although nine distinct attribute-concept clusters could be applied across these groups, the groups differed in what they emphasized most. Accordingly, they recommended that “a broad spectrum of personality concepts should be included in the development of common personality models and measurement tools for diverse cultural groups” (p. 365), with this implying an expansion of the Big Five set of constructs to their local context.

We find such studies that describe local and potentially emic personality constructs, while relating them both empirically and descriptively to outside frameworks, particularly useful and inspiring. In addition to their scientific value to an international audience, these kinds of studies can support the indigenization of psychology to new contexts (Adair, 2009), which emphasizes developing local knowledge that is adapted to local conditions and needs. Work by local researchers or intensive collaborations that substantively involve and engage researchers from under-represented societies, as in these examples, can support such goals.

Personality Psychology in Steven Heine’s *Cultural Psychology*

Another useful reference point for summarizing current views in cultural psychology is Heine’s (2020) popular textbook, from which many students first encounter this important aspect of psychology. This text includes many citations to work published in *JCCP* and it is representative of the kind of research emphasized in *JCCP* and conducted by IACCP members, including the content clusters cited by Gabrenya and Glazer (2022). Overall, we find it to be an excellent text that students enjoy reading, and which can inspire long-term interest in the topic. However, we find the coverage of personality traits rather mixed. After introducing the Big Five as the most widely accepted model among personality psychologists, Heine writes that, “The five-factor model proposes that all personality characteristics reflect some combination of the five core traits” (p. 239). This claim that the Big Five encompass all possible personality variation goes beyond even the most extravagant statements of Big Five advocates. As we have pointed out, the factors of a typical lexical study only account for roughly 25% of the variance. Recent high-dimensionality studies (e.g., Saucier & Iurino, 2020) show how much more variation there actually is. Similarly, Heine states that the Big Five are fundamental because they can’t be reduced to a smaller number and are largely uncorrelated with one another, which is contrary to substantial evidence for higher-order factors (DeYoung, 2006; Digman, 1997) and is furthermore not more true for the Big Five than for models with more factors (Ashton et al., 2004; Saucier & Iurino, 2020).

Given the proposition that personality can be accounted for by five factors, Heine addresses the question of cross-cultural generalization, asking the key question about whether it reflects “ideas about personhood that are limited to the West, where the bulk of this research has been conducted” (p. 240). The criterion he highlights, of whether it captures all the personality variation found worldwide, still starts with the assumption that all Big Five traits, including Openness and Extraversion, are found everywhere, and thus we might only need to add traits found in other places.

Finally, Heine cites research in a non-industrialized society (Gurven et al., 2013) using an interview measure of Big Five, which found little support for the structure, and he concludes that more research with other subsistence societies using different methods is needed before drawing conclusions about the generalizability of the five-factor model outside of WEIRD societies. We agree with that caution, but question its implication that the model is a settled matter within WEIRD societies, a view that we think the cross-cultural evidence does not support.

The Next 50 Years: How to Best Build on This Foundation?

With the benefit of hindsight, we take this opportunity to envision a future that successfully integrates the study of personality with attention to cultural context. We consider how universal

models and elements might be sought in more culturally-neutral ways, and propose a research agenda for lexical studies, noting the ways these can teach us about both personality and culture, and how they might ultimately lead to the identification of more truly universal models or frameworks. We consider recent examples of local research efforts focused on applied needs for personality assessment, and we join a growing movement in advocating for researcher reflexivity to bring more attention to the implicit assumptions and methodological choices that could impact study results.

If Not the Big Five: Could There Be Another Universal Structural Model?

For structures of one, two, and possibly three factors, evidence of universal salience seems clearer than for the Big Five or HEXACO/Big Six. For example, one-factor models contrasting socially desirable and undesirable traits (Saucier, Thalmayer, & Bel-Bahar, 2014), show an impressive degree of convergence across contexts (e.g., Thalmayer, Job et al., 2020). Similarly, van der Linden et al. (2018) confirmed the existence of a general factor of personality that contrasted socially-desirable with less-desirable attributes even among Amazonian forager-farmers, although this result came from the scales and items of a translated Big Five measure.

Two dimensions may also be highly replicable. Two-factor results from a globally diverse sample of studies supported a “Big Two” model, with factors of Social Self-Regulation and Dynamism (Saucier, Thalmayer, Payne, et al., 2014). Alternatively, De Raad et al. (2014) jointly analyzed lexical data from 10 European languages plus Filipino, and identified a three-factor solution with dimensions summarized as Dynamism, Affiliation, and Order as the core of the taxonomies. These parsimonious models may have better tractability for being accounted for theoretically, and their cross-culturally generalizability could indicate either a genetic or biological basis for the attributes included, or their basis in universal nonbiological features characterizing human social environments (Fontaine, 2011; Saucier, 2009).

There are two possible approaches to defining a universal model. First, one can seek the replication of a set of factors, as has been assumption in the work on the Big Five. It is on these grounds that Big Two appears to be a stronger model (Saucier, Thalmayer, & Bel-Bahar, 2014). Alternatively, however, a large set of granular constructs could be considered, and specific traits can be assessed for replication and consistently, without demanding that the whole set match. This approach was used in a study of 12 languages from diverse social and ecological contexts which were unlikely to have influenced each other (Saucier, Thalmayer, & Bel-Bahar, 2014). These approaches can potentially be complementary. Models that meet the first, higher criteria are likely to include only a few broad dimensions and thus to lack predictive power, but can be used to organize more elaborated, high-dimensional results.

Structures of just one, two, or three dimensions are of course far from comprehensive of all the variation among personality traits. Even five-factor models, however, do not provide “a comprehensive mapping of personality traits” (contrary to Ispas et al., 2014, p. 1074). Recent re-analyses, in an era of greater computational power, indicate that emic lexical-study results have much higher potential dimensionality (Saucier & Iurino, 2020). If a dozen or more factors can be identified in each language-community, a promising approach could be to compare the emic-emergent factors from a large number of languages in a shared matrix. Of dozens of possible dimensions, some will be more emphasized in some communities, indicating local patterns of concern and interest. The contents could be ranked from most universally salient (appearing in all the languages), to more moderately universal, to emic (specific to a single context). Any model with five (or six) factors, each of which is a kind of mosaic of multiple granular elements, will fail to replicate perfectly everywhere because of the potential for languages to include or exclude various parts of the five (or six) mosaics of personality traits in different ways.

This would have analogies to the study of emotion (also, to a degree, psychopathology), which has a long and useful tradition of studying many fine-grained constructs, and thereby concluding for example that anger is a more universal concept than *Schadenfreude*. In this way, personality psychology might better balance the benefits of a common model with the potentially complex diversity of indigenous constructs. So far, the discovery of culture-specific personality traits has been limited by the predominant use of five broad constructs with somewhat fuzzy boundaries. For example, someone trained only in the five-or-so factors method paradigm may miss culture-specific traits, because constraining factors to a few makes them more generic collages, and within the collage one may usually find *something* that seems to fit.

A Research Agenda for Lexical Studies of Personality

An underappreciated aspect of lexical studies is their potential to reveal important aspects of local societies. Factor analysis does not only identify which terms correlate with each other, but also which factors best distinguish between cases in the data. The identification of which person-descriptor terms are most frequently used among a cultural group, and how they fit together, reveals local patterns of discourse about character. For example, in Goldberg's original English-language study, Extraversion was a large factor (1990): A rich vocabulary related to this domain allows North Americans to make subtle distinctions, and the visibility of this trait makes variation in scores unsurprising. But few terms related to Extraversion appear among commonly-used person-descriptive terms in three representative African languages (Thalmayer, Job et al., 2020; Thalmayer, Saucier et al., 2020). This domain is clearly important in the United States where people move often and interact with many strangers, creating high "relational mobility" (Zhang & Li, 2014) and situations in which a person's degree of Extraversion is apparent. Depue and Collins (1999) defined the core mechanism of this trait to be reward seeking. The importance of this trait, demonstrated by how much attention those in the United States bring to it, could also be said to illustrate a key feature of American society, where rewards are sought in a highly competitive, individualistic, high income-inequality context (Nelson & Shavitt, 2002).

Speakers of Supyire-Senufo in West Africa have a different lifestyle, commonly staying in the same rural area throughout their lives, sharing the labor of subsistence farming within stable communities. There, a trait dimension of "laziness" was highly salient; their most-used terms included many that defined subtle nuances in industriousness, but almost none related to Extraversion. Extraversion has not been as useful to talk about there. In Khoekhoegowab, a language spoken among two Southern African groups who were traditionally herders or hunter-gatherers, then typically farm workers in colonial times, and who now live in small towns and villages in a developing economy, none of the 272 most-used terms to describe people related to Extraversion. The first unrotated factor identified was instead one of "Temperance," contrasting being religious with alcohol use (Thalmayer, Job et al., 2020). Later work among Khoekhoegowab-speakers on the topic of mental health helped to elucidate this distinction as contrasting two key means of coping with adversity and hardship, both involving the value of connecting oneself to a community, for example, the concept of *Khoesis* (or in nearby languages, *Ubuntu* or *Botho*; Sodi et al., 2021). Do you turn to prayer and community through religion, or do you turn to drink and community at the bar? In a context with minimal state or institutionalized services, the particular way a person contributes to the social bonds that make up the local safety net appear to form a more salient individual difference than do Extraversion or Intellect.

Recent methodological and conceptual innovations in lexical studies of personality, which bring more attention to the identification of an optimal local model and make a clearer distinction between emic and etic analyses, have helped highlight such cultural differences. The use of parallel analysis and systematic comparisons of analytic/methodological variation to identify optimal local models has allowed for identification of larger numbers of factors that are both interpretable

and replicable (Saucier & Iurino, 2020; Thalmayer, Job et al., 2020; Thalmayer, Saucier et al., 2020). These studies have welcomed complexity and local nuance rather than emphasizing parsimony in the structure of individual differences, reporting on many constructs instead of only a few. This resonates with the goals of other contemporary personality psychologists, who point out that more specific facets have more predictive power when disaggregated from models like the Big Five (Möttus, 2016).

While personality studies are primarily intended to capture differences between individuals in a society, lexical studies can thus be conducted in a way that is also relevant to cultural psychology, highlighting local values, and concerns. Methodical comparison of results to a background framework of granular attribute-concepts could enable future studies to discern between what is culturally specific, minimally available, or universally salient. Moreover, these studies could be extended to capture cultural values by assessing the desirability of the attribute-concepts. Such values, as well as aspects of attribute structure, can be compared between subpopulations, for example, educated versus uneducated. These innovations could make lexical studies more appealing to researchers throughout the world, including interdisciplinary teams: such studies can reveal local preoccupations and explore how locally salient personality descriptors converge with findings from social, developmental, and clinical psychology, as well as from history, sociology, and cultural studies.

Applied Personality Assessment Across Cultures

Ambitious emic projects have led to the creation of local personality inventories, for example the Chinese Personality Assessment Inventory, described above. Similar projects have been underway in South Africa (Fetvadjev et al., 2015; Nel et al., 2012) and with speakers of Arabic in the Levant (Zeinoun et al., 2017), where methods have been tailored to local circumstances. For example, for the South Africa Personality Inventory (SAPI), the context of having 11 official languages led to a strategy of using free personality descriptions in each language, followed by qualitative content analysis to identify clusters, then defined as traits to be measured with short phrases. The initial qualitative work included over 1,200 semi-structured interviews in 11 languages (Nel et al., 2012), a heroic effort that led to 37 subclusters merged into nine broad clusters. For these, 2,268 valid and reliable items were developed (Hill et al., 2013). In Arabic, similar qualitative analyses of over 17,000 open-ended descriptions in multiple areas and vernaculars of spoken Arabic led to a model of nine broad clusters, later converted into a seven-factor instrument. In both cases, local models were found to be more elaborated than the Big Five, but to include most Big Five content, though this was weakest for Openness.

These projects have the important strength of drawing on common phrasing, and of capturing personality description in a more contextual way than single words. This maximizes the value of these inventories for local, applied use, and, in the process, has led to the creation of valuable corpora of personality description, which could be useful for many future research purposes. Lexical studies, which take a more exhaustive and conservative starting point with a published dictionary, include only content that has been in the language for some time and the words are separated from context when administered. An ongoing challenge in a more open-ended approach, however, may be in defining the final model. The currently used SAPI includes only six of the initial nine factors, basically the Big Five with Agreeableness reconceptualized and divided into Positive and Negative-Social-Relational factors (Fetvadjev et al., 2015). It is a bit hard to imagine the comparatively weak dimensions of Openness and Extraversion being included in the final model without the strong priming influence of the Big Five, and priming may also be implicated in the choice of how many factors to include in final inventories. This illustrates how cross-cultural work on personality structure shapes applied work. The identification of a more universal model could facilitate such projects, ideally offering a common higher-order structure for comparison purposes, while also allowing

for more elaborated, locally-adapted models to maximize prediction and utility. We look forward to the future developments of this impressive inclusionary and exploratory project in South Africa, and to the ways it could inspire similar efforts in other contexts.

Lexical studies that identify local structures can also be a source for the development of locally-adapted questionnaires. A 38-item questionnaire based on key terms for 10 lexical factors in Khoekhoegowab, for example, was shown to out-perform a Big Six inventory in predicting mental health and religious engagement (Thalmayer et al., 2021). In addition to predictive validity, it could be useful to compare local and imported questionnaires in terms of capturing the variation within subjects, as explored by Allik et al. (2018).

Researcher Reflexivity and Broader Questions about Personality Psychology

While discussions of ontology and reflexivity (ongoing evaluation of one's role in the research process and subjective experiences) are more associated with qualitative research, they have great potential value for quantitative psychologists as well (Finlay, 1998; Jafar, 2018). As described above, some choices in lexical studies may have inadvertently played a role in exaggerating cross-cultural support for the Big Five. And imposed-etic questionnaire studies are implicitly based on the assumption that personality structure, specifically the Five Factor Model, is a feature of human brains that is not shaped by the social and cultural context (explicitly stated by McCrae & Costa PT, 1997). It also implies a positivist ontological position that reality and the social world exist independently of the observer's perception (Hudson & Ozanne, 1988), and that the world corresponds to permanent rules of causation and occurrences (Aliyu et al., 2014). Such assumptions influence choices in personality research, which may be treated as more objective than they really are. Five Factor Theory is also at odds with overwhelming evidence that the cultural context profoundly impacts the brain, biology, and psychology (e.g., Heine, 2020; Henrich, 2020; Henrich et al., 2022).

Cross-cultural research inevitably requires careful attention to methodology and reflection on our assumptions, making researcher self-reflection appropriate and useful. Furthermore, there are many advantages to including qualitative and mixed-methods aspects to studies in contexts where there is scant prior research. Thus, we believe that Levitt et al. (2018) useful recommendations to qualitative researchers may generalize here. These include being aware of the researcher's influence (their expectations, values, or assumptions) on the analytic questions and the interpretation of results, considering the context within which the study is conducted, the relationship of the researcher to the study topic and the relation to social dynamics, being transparent in reporting data collection and analysis choices, and reflecting on alternative explanations of the findings (Levitt et al., 2018).

An interesting question related to such self-reflection is the extent to which personality differences are more apparent and, even, interesting in certain kinds of cultural contexts, namely Western, individualistic ones. Although the patterns described as personality suggest social effects and social relations, the focus on a person as a unit separate from others with a distinct set of properties is an individualistic starting point (Saucier, in press; Henrich, 2020). Cross-cultural differences have been seen in the extent to which personality attributes are seen as fixed and enduring, versus malleable and influenced by the situation (Heine, 2012; Kanagawa et al., 2001). In individualistic cultures, it is considered desirable to have a consistent identity across situations and personality consistency is associated with social skills, being well-liked, and well-being (Suh, 2002). In collectivistic cultures, being consistent across situations is not associated with such rewards and a higher value is placed on the social skills involved in appropriate adaptation to one's roles (Suh, 2002). Cultural differences in "dispositionalism" (Henrich, 2020), the degree of the perceived relevance of the personal attributes of individuals, are worth considering in and of themselves, as well as in terms of their impact on personality assessment and research. One useful direction might be the Culture $\times$ Person $\times$ Situation approach proposed by Leung and Cohen (2011), to seek integrated accounts of within- and between-culture variation.

Conclusions

Human beings exist within cultures, leading individual personalities to be interwoven with cultural influences. While words to define personality characteristics appear in all languages, indicating the universal relevance of individual differences, the stable psychological patterns that can be defined as traits are experienced within a cultural matrix that differentially permits, celebrates, and notices them. While studying personality and culture together presents many challenges, the study of personality is incomplete without consideration of context.

As is true for research in all domains of psychology over the last 50 years, the study of personality has sometimes tended to over-impose Western models, with too little emphasis on exploring diversity or identifying potential universals in non-Western contexts. Personality psychology has, however, also included early, laudable attempts to integrate perspectives from diverse cultures. Personality research in the *JCCP* using nuanced approaches to testing etic models in new contexts, or integrating emic and etic approaches, provide good examples for the future study of personality across cultures. We hope that a new generation of researchers from around the world will be inspired by the foundational and exploratory work of the last 50 years to creatively face the challenge of integrating the study of personality with culture.

Important questions that remain to be addressed include the identification of more universal structural models. We join Adamopoulos and Lonner (1994), Fontaine (2011), and Heine (2020) in recommending that researchers be precise in their definitions of what is more or less universal, and we advocate for reflexivity about the role of the researcher, and pre-registered analysis plans with explicit etic/emic strategies to seek both universal and culturally-specific elements. The future of cross-cultural personality research will benefit from curiosity regarding the nuances of traits in diverse settings. In this way, the study of personality traits can teach us about human societies as well as about individuals.

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Note

1. A fourth category of purely emic or indigenous studies was defined by Cheung et al. (2011), including bottom-up approaches to examine culture-specific phenomena. They use lexical studies as an example, but as described above, we find that this depends a great deal on how the study is conducted. Given the cross-cultural emphasis of *JCCP*, it is perhaps not surprising that we identified few articles here that seemed best described by this fourth category. While some put their emic portions more at the forefront, *JCCP* personality articles generally seek to integrate local constructs with existing models, at least descriptively.

References

- Adair, J. G. (2009). Indigenization and beyond: The process and extent of psychology's growth as an international science. In S. Bekman & A. Aksu-Koç (Eds.), *Perspectives on human development, family, and culture* (pp. 3–16). Cambridge University Press.

- Adamopoulos, J., & Lonner, W. J. (1994). Absolutism, relativism, and universalism in the study of culture and human activity. In W. J. Lonner & R. Malpass (Eds.), *Psychology and culture* (pp. 129–134). Allyn and Bacon.
- Adams, G., Dobles, I., Gómez, L. H., Kurtiş, T., & Molina, L. E. (2015). Decolonizing psychological science: Introduction to the special thematic section. *Journal of Social and Political Psychology*, 3(1), 213–238. <https://doi.org/10.5964/jssp.v3i1.564>
- Aliyu, A. A., Bello, M. U., Kasim, R., & Martin, D. (2014). Positivist and non-positivist paradigm in social science research: Conflicting paradigms or perfect partners? *Journal of Management and Sustainability*, 4(3), 79–95. <https://doi.org/10.5539/jms.v4n3p79>
- Allik, J., Church, A. T., Ortiz, F. A., Rossier, J., Hřebíčková, M., de Fruyt, F., Realo, A., & McCrae, R. R. (2017). Mean profiles of the NEO Personality Inventory. *Journal of Cross-Cultural Psychology*, 48(3), 402–420. <https://doi.org/10.1177/0022022117692100>
- Allik, J., Hřebíčková, M., & Realo, A. (2018). Unusual configurations of personality traits indicate multiple patterns of their coalescence. *Frontiers in Psychology*, 9(187), 187–214. <https://doi.org/10.3389/fpsyg.2018.00187>
- Allik, J., & McCrae, R. R. (2004). Toward a geography of personality traits: Patterns of profiles across 36 cultures. *Journal of Cross-Cultural Psychology*, 35(1), 13–28. <https://doi.org/10.1177/0022022103260382>
- Allik, J., & Realo, A. (2017). Universal and specific in the five-factor model of personality. In T. A. Widiger (Ed.), *The Oxford handbook of the five-Factor Model of personality* (pp. 175–190). Oxford University Press.
- Allik, J., Realo, A., Möttus, R., Pullmann, H., Trifonova, A., McCrae, R. R., Yurina, A. A., Shebanets, E. Y., Fadina, A. G., Tikhonova, E. V., Troitskaya, I. Y., Fatyhova, R. M., Petrova, I. F., Tudupova, T. T., Tsiring, D. A., Panfilov, A. N., L'dokova, G. M., Aristova, I. L., Mescheryakov, B. G., . . . Korneeva, E. E. (2011). Personality profiles and the “Russian soul”: Literary and scholarly views evaluated. *Journal of Cross-Cultural Psychology*, 42(3), 372–389. <https://doi.org/10.1177/0022022110362751>
- Allport, G. W., & Odbert, H. S. (1936). Trait-names: A psycho-lexical study. *Psychological Monographs*, 47(1), i–171. <https://doi.org/10.1037/h0093360>
- Angleitner, A., Ostendorf, F., & John, O. P. (1990). Towards a taxonomy of personality descriptors in German: A psycho-lexical study. *European Journal of Personality*, 4(2), 89–118. <https://doi.org/10.1002/per.2410040204>
- Ashton, M. C., Lee, K., & Goldberg, L. R. (2004). A hierarchical analysis of 1,710 English personality-descriptive adjectives. *Journal of Personality and Social Psychology*, 87(5), 707–721. <https://doi.org/10.1037/0022-3514.87.5.707>
- Bartram, D. (2013). Scalar equivalence of OPQ32: Big five profiles of 31 countries. *Journal of Cross-Cultural Psychology*, 44(1), 61–83. <https://doi.org/10.1177/0022022111430258>
- Baumgarten, F. (1933). *Die Charaktereigenschaften. Beiträge zur Charakter- und Persönlichkeitsforschung. Monogr. I.* . A. Francke.
- Behrens, K. Y. (2004). A multifaceted view of the concept of Amae: Reconsidering the indigenous Japanese concept of relatedness. *Human Development*, 47(1), 1–27. <https://doi.org/10.1159/000075366>
- Bijnen, E. J., & Poortinga, Y. H. (1988). The questionable value of cross-cultural comparisons with the Eysenck Personality Questionnaire. *Journal of Cross-Cultural Psychology*, 19(2), 193–202. <https://doi.org/10.1177/0022022188192005>
- Bijnen, E. J., Van Der Net, T. Z. J., & Poortinga, Y. H. (1986). On cross-cultural comparative studies with the Eysenck Personality Questionnaire. *Journal of Cross-Cultural Psychology*, 17(1), 3–16. <https://doi.org/10.1177/0022002186017001001>
- Caprara, G. V., Barbaranelli, C., Bermúdez, J., Maslach, C., & Ruch, W. (2000). Multivariate methods for the comparison of factor structures in cross-cultural research: An illustration with the Big Five questionnaire. *Journal of Cross-Cultural Psychology*, 31(4), 437–464. <https://doi.org/10.1177/0022022100031004002>
- Cheung, F. M., Cheung, S. F., Leung, K., Ward, C., & Leong, F. (2003). The English version of the Chinese Personality Assessment Inventory. *Journal of Cross-Cultural Psychology*, 34(4), 433–452. <https://doi.org/10.1177/0022022103034004004>
- Cheung, F. M., Cheung, S. F., Zhang, J., Leung, K., Leong, F., & Huiyeh, K. (2008). Relevance of openness as a personality dimension in Chinese culture: Aspects of its cultural relevance. *Journal of Cross-Cultural Psychology*, 39(1), 81–108. <https://doi.org/10.1177/0022022107311968>

- Cheung, F. M., & Leung, K. (1998). Indigenous personality measures: Chinese examples. *Journal of Cross-Cultural Psychology*, 29(1), 233–248. <https://doi.org/10.1177/0022022198291012>
- Cheung, F. M., Leung, K., Fan, R. M., Song, W.-Z., Zhang, J.-X., & Zhang, J.-P. (1996). Development of the Chinese Personality Assessment Inventory. *Journal of Cross-Cultural Psychology*, 27(2), 181–199. <https://doi.org/10.1177/0022022196272003>
- Cheung, F. M., Leung, K., Zhang, J.-X., Sun, H.-F., Gan, Y.-Q., Song, W.-Z., & Xie, D. (2001). Indigenous Chinese personality constructs: Is the five-factor model complete? *Journal of Cross-Cultural Psychology*, 32(4), 407–433. <https://doi.org/10.1177/0022022101032004003>
- Cheung, F. M., van de Vijver, F. J. R., & Leong, F. T. L. (2011). Toward a new approach to the study of personality in culture. *American Psychologist*, 66(7), 593–603. <https://doi.org/10.1037/a0022389>
- Church, A. T., & Katigbak, M. S. (1988). The emic strategy in the identification and assessment of personality dimensions in a non-Western culture: Rationale, steps, and a philippine illustration. *Journal of Cross-Cultural Psychology*, 19(2), 140–163. <https://doi.org/10.1177/0022022188192002>
- Church, A. T., & Lonner, W. J. (1998). The cross-cultural perspective in the study of personality: Rationale and current research. *Journal of Cross-Cultural Psychology*, 29(1), 32–62. <https://doi.org/10.1177/0022022198291003>
- Church, A. T., Reyes, J. A., Katigbak, M. S., & Grimm, S. D. (1997). Filipino personality structure and the Big Five model: A lexical approach. *Journal of Personality*, 65(3), 477–528. <https://doi.org/10.1111/j.1467-6494.1997.tb00325.x>
- Depue, R. A., & Collins, P. F. (1999). Neurobiology of the structure of personality: Dopamine, facilitation of incentive motivation, and extraversion. *Behavioral and Brain Sciences*, 22(3), 491–517; discussion 518. <https://doi.org/10.1017/S0140525X99252041>
- De Raad, B., Barelds, D. P., Levert, E., Ostendorf, F., Mlacić, B., Di Blas, L., Hrebícková, M., Szirmák, Z., Szarota, P., Perugini, M., Church, A. T., & Katigbak, M. S. (2010). Only three factors of personality description are fully replicable across languages: A comparison of 14 trait taxonomies. *Journal of Personality and Social Psychology*, 98(1), 160–173. <https://doi.org/10.1037/a0017184>
- De Raad, B., Barelds, D. P. H., Timmerman, M. E., De Roover, K., Mlacić, B., & Church, A. T. (2014). Towards a pan-cultural personality structure: Input from 11 psycholexical studies. *European Journal of Personality*, 28(5), 497–510. <https://doi.org/10.1002/per.1953>
- De Raad, B., Hendriks, A. A. J., & Hofstee, W. K. B. (1992). Towards a refined structure of personality traits. *European Journal of Personality*, 6(4), 301–319. <https://doi.org/10.1002/per.2410060405>
- De Raad, B., Perugini, M., Hrebícková, M., & Szarota, P. (1998). Lingua franca of personality: Taxonomies and structures based on the psycholexical approach. *Journal of Cross-Cultural Psychology*, 29(1), 212–232. <https://doi.org/10.1177/0022022198291011>
- DeYoung, C. G. (2006). Higher-order factors of the Big Five in a multi-informant sample. *Journal of Personality and Social Psychology*, 91(6), 1138–1151. <https://doi.org/10.1037/0022-3514.91.6.1138>
- Digman, J. M. (1997). Higher-order factors of the Big Five. *Journal of Personality and Social Psychology*, 73(6), 1246–1256. <https://doi.org/10.1037/0022-3514.73.6.1246>
- Dixon, R. M. (1982). *Where have all the adjectives gone?: And other essays in semantics and syntax*. Walter de Gruyter.
- Eigenhuis, A., Kamphuis, J. H., & Noordhof, A. (2015). Personality differences between the United States and the Netherlands: The influence of violations of measurement invariance. *Journal of Cross-Cultural Psychology*, 46(4), 549–564. <https://doi.org/10.1177/0022022115570671>
- Fetvadjev, V. H., Meiring, D., van de Vijver, F. J., Nel, J. A., & Hill, C. (2015). The South African Personality Inventory (SAPI): A culture-informed instrument for the country's main ethnocultural groups. *Psychological Assessment*, 27(3), 827–837. <https://doi.org/10.1037/pas0000078>
- Finlay, L. (1998). Reflexivity: An essential component for all research? *British Journal of Occupational Therapy*, 61(10), 453–456. <https://doi.org/10.1177/030802269806101005>
- Fischer, R., & Karl, J. A. (2019). A primer to (cross-cultural) multi-group invariance testing possibilities in R. *Frontiers in Psychology*, 10, 1507. <https://doi.org/10.3389/fpsyg.2019.01507>
- Fontaine, J. R. J. (2011). A fourfold conceptual framework for cultural and cross-cultural psychology: Relativism, construct universalism, repertoire universalism and absolutism. In F. J. R. van de Vijver, A. Chasiotis, & S. M. Breugelmans (Eds.), *Fundamental questions in cross-cultural psychology* (pp. 165–189). Cambridge University Press.

- Funder, D. C. (2001). Accuracy in personality judgment: Research and theory concerning an obvious question. In B. W. Roberts & R. Hogan (Eds.), *Personality psychology in the workplace* (pp. 121–140). American Psychological Association.
- Gabrenya, W., & Glazer, S. (2022). Bridging 50 years of theoretical and applied cross-cultural psychology: Contributions of IACCP and JCCP. *Journal of Cross-Cultural Psychology*, 53(7-8), 752–788.
- Galtung, J. (1967). Scientific colonialism. *Transition*, 30, 10–15. <https://doi.org/10.2307/2934342>
- Goldberg, L. (1981). Language and individual differences: The search for universals in personality lexicons. In L. Wheeler (Ed.), *Review of personality and Social Psychology* (pp. 141–165). SAGE Publication.
- Goldberg, L. R. (1990). An alternative “description of personality”: The Big-Five factor structure. *Journal of Personality and Social Psychology*, 59(6), 1216–1229. <https://doi.org/10.1037/0022-3514.59.6.1216>
- Goldberg, L. R., & Somer, O. (2000). The hierarchical structure of common Turkish person-descriptive adjectives. *European Journal of Personality*, 14(6), 497–531. [https://doi.org/10.1002/1099-0984\(200011/12\)14:6<497::aid-per379>3.0.co;2-r](https://doi.org/10.1002/1099-0984(200011/12)14:6<497::aid-per379>3.0.co;2-r)
- Guanzon-Lapeña, M. A., Church, A. T., Carlota, A. J., & Katigbak, M. S. (1998). Indigenous personality measures: philippine examples. *Journal of Cross-Cultural Psychology*, 29(1), 249–270. <https://doi.org/10.1177/0022022198291013>
- Gurven, M., von Rueden, C., Massenkoff, M., Kaplan, H., & Lero Vie, M. (2013). How universal is the Big Five? Testing the five-factor model of personality variation among forager-farmers in the Bolivian Amazon. *Journal of Personality and Social Psychology*, 104(2), 354–370. <https://doi.org/10.1037/a0030841>
- Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2017). *Multivariate data analysis* (7th ed.). Pearson.
- Heine, S. J. (2012). *Cultural psychology* (2nd ed.). W. W. Norton & Company.
- Heine, S. J. (2020). *Cultural psychology* (4th ed.). W. W. Norton & Company.
- Henrich, J. (2016). *The secret of our success: How culture is driving human evolution, domesticating our species, and making us smarter*. Princeton University Press.
- Henrich, J. (2020). *The WEIRDest people in the world: How the West became psychologically peculiar and particularly prosperous*. Penguin UK.
- Henrich, J., Blasi, D. E., Curtin, C. M., Davis, H. E., Hong, Z., Kelly, D., & Kroupin, I. (2022). A cultural species and its cognitive Phenotypes: Implications for Philosophy. *Review of Philosophy and Psychology*. Advance online publication. <https://doi.org/10.1007/s13164-021-00612-y>
- Henrich, J., Heine, S. J., & Norenzayan, A. (2010). The weirdest people in the world? *Behavioral and Brain Sciences*, 33(2-3), 61–83; discussion 83. <https://doi.org/10.1017/S0140525X0999152X>
- Hill, C., Nel, J. A., Van de Vijver, F. J. R., Meiring, D., Valchev, V. H., Adams, B. G., & De Bruin, G. P. (2013). Developing and testing items for the South African Personality Inventory (SAPI). *SA Journal of Industrial Psychology/SA Tydskrif vir Bedryfsielkunde*, 39(1), 1–13. <https://doi.org/10.4102/sajip.v39i1.1122>
- Hopwood, C. J., & Donnellan, M. B. (2010). How should the internal structure of personality inventories be evaluated? *Personality and Social Psychology Review*, 14(3), 332–346. <https://doi.org/10.1177/1088868310361240>
- Horn, J. L. (1967). On subjectivity in factor analysis. *Educational and Psychological Measurement*, 27(4), 811–820. <https://doi.org/10.1177/001316446702700404>
- Huang, C. D., Church, A. T., & Katigbak, M. S. (1997). Identifying cultural differences in items and traits: Differential item functioning in the NEO Personality Inventory. *Journal of Cross-Cultural Psychology*, 28(2), 192–218. <https://doi.org/10.1177/0022022197282004>
- Hudson, L. A., & Ozanne, J. L. (1988). Alternative ways of seeking knowledge in consumer research. *Journal of Consumer Research*, 14(4), 508. <https://doi.org/10.1086/209132>
- Ispas, D., Iliescu, D., Ilie, A., & Johnson, R. E. (2014). Exploring the cross-cultural generalizability of the five-factor model of personality: The Romanian NEO PI-R. *Journal of Cross-Cultural Psychology*, 45(7), 1074–1088. <https://doi.org/10.1177/0022022114534769>
- Jafar, A. J. N. (2018). What is positionality and should it be expressed in quantitative studies? *Emergency Medicine Journal*, 35(5), 323–324. <https://doi.org/10.1136/emered-2017-207158>
- Kaiser, H. F. (1960). The application of electronic computers to factor analysis. *Educational and Psychological Measurement*, 20, 141–151. <https://doi.org/10.1177/001316446002000116>

- Kanagawa, C., Cross, S. E., & Markus, H. R. (2001). "Who am I?" the cultural psychology of the conceptual self. *Personality and Social Psychology Bulletin*, 27(1), 90–103. <https://doi.org/10.1177/0146167201271008>
- Kim, H., Schimmack, U., Oishi, S., & Tsutsui, Y. (2018). Extraversion and life satisfaction: A cross-cultural examination of student and nationally representative samples. *Journal of Personality*, 86(4), 604–618. <https://doi.org/10.1111/jopy.12339>
- Laajaj, R., Macours, K., Pinzon Hernandez, D. A., Arias, O., Gosling, S. D., Potter, J., Rubio-Codina, M., & Vakis, R. (2019). Challenges to capture the Big Five personality traits in non-weird populations. *Science Advances*, 5(7), 1–13. <https://doi.org/10.1126/sciadv.aaw5226>
- Lee, K., & Ashton, M. C. (2009). Reanalysis of the structure of the Greek personality lexicon. *Journal of Cross-Cultural Psychology*, 40(4), 693–700. <https://doi.org/10.1177/0022022109335183>
- Lefley, H. P. (1972). Modal personality in the Bahamas. *Journal of Cross-Cultural Psychology*, 3(2), 135–147. <https://doi.org/10.1177/002202217200300202>
- Leung, A. K., & Cohen, D. (2011). Within- and between-culture variation: Individual differences and the cultural logics of honor, face, and dignity cultures. *Journal of Personality and Social Psychology*, 100(3), 507–526. <https://doi.org/10.1037/a0022151>
- Levitt, H. M., Bamberg, M., Creswell, J. W., Frost, D. M., Josselson, R., & Suárez-Orozco, C. (2018). Journal article reporting standards for qualitative primary, qualitative meta-analytic, and mixed methods research in psychology: The APA Publications and Communications Board task force report. *American Psychologist*, 73(1), 26–46. <https://doi.org/10.1037/amp0000151>
- Lin, E. J.-L., & Church, A. T. (2004). Are indigenous Chinese personality dimensions culture-specific?: An investigation of the Chinese Personality Assessment Inventory in Chinese American and European American samples. *Journal of Cross-Cultural Psychology*, 35(5), 586–605. <https://doi.org/10.1177/0022022104268390>
- MacDonald, K. (1998). Evolution, culture, and the five-factor model. *Journal of Cross-Cultural Psychology*, 29(1), 119–149. <https://doi.org/10.1177/0022022198291007>
- McCrae, R. R., & Costa, P. T. (1992). Discriminant validity of NEO-PIR facet scales. *Educational and Psychological Measurement*, 52(1), 229–237. <https://doi.org/10.1177/001316449205200128>
- McCrae, R. R., & Costa, P. T., Jr. (1997). Personality trait structure as a human universal. *American Psychologist*, 52(5), 509–516. <https://doi.org/10.1037/0003-066X.52.5.509>
- Möttus, R. (2016). Towards more rigorous personality trait–outcome research. *European Journal of Personality*, 30(4), 292–303. <https://doi.org/10.1002/per.2041>
- Möttus, R., Allik, J., Realo, A., Pullmann, H., Rossier, J., Zecca, G., Ah-Kion, J., Amoussou-Yéyé, D., Bäckström, M., Barkauskiene, R., Barry, O., Bhowon, U., Björklund, F., Bochaver, A., Bochaver, K., de Bruin, G. P., Cabrera, H. F., Chen, S. X., Church, A. T., . . . , Ng Tseung, C. (2012). Comparability of self-reported conscientiousness across 21 countries. *European Journal of Personality*, 26(3), 303–317. <https://doi.org/10.1002/per.840>
- Nel, J. A., Valchev, V. H., Rothmann, S., Vijver, F. J. R., Meiring, D., & Bruin, G. P. (2012). Exploring the personality structure in the 11 languages of South Africa: Personality structure in South Africa. *Journal of Personality*, 80(4), 915–948. <https://doi.org/10.1111/j.1467-6494.2011.00751.x>
- Nelson, M. R., & Shavitt, S. (2002). Horizontal and vertical individualism and achievement values: A multimethod examination of Denmark and the United States. *Journal of Cross-Cultural Psychology*, 33(5), 439–458. <https://doi.org/10.1177/0022022102033005001>
- Norenzayan, A., & Heine, S. J. (2005). Psychological universals: What are they and how can we know? *Psychological Bulletin*, 131(5), 763–784.
- Paunonen, S. V., & Ashton, M. C. (1998). The structured assessment of personality across cultures. *Journal of Cross-Cultural Psychology*, 29(1), 150–170. <https://doi.org/10.1177/0022022198291008>
- Piedmont, R. L., & Chae, J.-H. (1997). Cross-cultural generalizability of the five-factor model of personality: Development and validation of the NEO PI-R for Koreans. *Journal of Cross-Cultural Psychology*, 28(2), 131–155. <https://doi.org/10.1177/0022022197282001>
- Saab, R., Ayanian, A. H., & Hawi, D. R. (2020). The status of Arabic social psychology: A review of 21st-century research articles. *Social Psychological and Personality Science*, 11(7), 917–927. <https://doi.org/10.1177/1948550620925224>
- Saucier, G. (1997). Effects of variable selection on the factor structure of person descriptors. *Journal of Personality and Social Psychology*, 73(6), 1296–1312. <https://doi.org/10.1037/0022-3514.73.6.1296>

- Saucier, G. (2009). Recurrent personality dimensions in inclusive lexical studies: Indications for a big Six structure. *Journal of Personality*, 77(5), 1577–1614. <https://doi.org/10.1111/j.1467-6494.2009.00593.x>
- Saucier, G. (in press). *Culture, personality, and the psychology of religion*. Brill.
- Saucier, G., & Goldberg, L. R. (1998). What is beyond the Big Five? *Journal of Personality*, 66, 495–524. <https://doi.org/10.1111/1467-6494.00022>
- Saucier, G., & Iurino, K. (2020). High-dimensionality personality structure in the natural language: Further analyses of classic sets of English-language trait-adjectives. *Journal of Personality and Social Psychology*, 119(5), 1188–1219. <https://doi.org/10.1037/pspp0000273>
- Saucier, G., Thalmayer, A. G., & Bel-Bahar, T. S. (2014). Human attribute concepts: Relative ubiquity across twelve mutually isolated languages. *Journal of Personality and Social Psychology*, 107(1), 199–216. <https://doi.org/10.1037/a0036492>
- Saucier, G., Thalmayer, A. G., Payne, D. L., Carlson, R., Sanogo, L., Ole-Kotikash, L., Church, A. T., Katigbak, M. S., Somer, O., Szarota, P., Szirmák, Z., & Zhou, X. (2014). A basic bivariate structure of personality attributes evident across nine languages. *Journal of Personality*, 82(1), 1–14. <https://doi.org/10.1111/jopy.12028>
- Schmitt, D. P., Allik, J., McCrae, R. R., & Benet-Martínez, V. (2007). The geographic distribution of big five personality traits: Patterns and profiles of human self-description across 56 nations. *Journal of Cross-Cultural Psychology*, 38(2), 173–212. <https://doi.org/10.1177/0022022106297299>
- Sodi, T., Bopape, D., & Makgahlela, M. (2021). Botho as an essential ingredient of African psychology: An insider perspective. *South African Journal of Psychology*, 51(3), 441–452. <https://doi.org/10.1177/0081246320985259>
- Suh, E. M. (2002). Culture, identity consistency, and subjective well-being. *Journal of Personality and Social Psychology*, 83(6), 1378–1391. <https://doi.org/10.1037/0022-3514.83.6.1378>
- Tellegen, A., & Waller, N. G. (2008). Exploring personality through test construction: Development of the Multidimensional Personality Questionnaire. In G. J. Boyle, G. Matthews, & D. H. Saklofske (Eds.), *The SAGE handbook of personality theory and assessment, Vol. 2. Personality measurement and testing* (pp. 261–292). SAGE Publications, Inc.
- Thalmayer, A. G., Job, S., Shino, E. N., Robinson, S. L., & Saucier, G. (2020). †Ušigu: A mixed-method lexical study of character description in Khoekhoegowab. *Journal of Personality and Social Psychology*, 121(6), 1258–1283. <https://doi.org/10.1037/pspp0000372>
- Thalmayer, A. G., Saucier, G., Ole-Kotikash, L., & Payne, D. (2020). Personality structure in east and west Africa: Lexical studies of personality in Maa and Supyire-Senufo. *Journal of Personality and Social Psychology*, 119(5), 1132–1152. <https://doi.org/10.1037/pspp0000264>
- Thalmayer, A. G., Saucier, G., Shino, E. N., & Job, S. (2021). The Khoekhoegowab Personality Inventory: The comparative validity of a locally derived measure of traits. *Frontiers in Psychology*, 12, 694205. <https://doi.org/10.3389/fpsyg.2021.694205>
- Valchev, V. H., Nel, J. A., van de Vijver, F. J. R., Meiring, D., de Bruin, G. P., & Rothmann, S. (2013). Similarities and differences in implicit personality concepts across ethnocultural groups in South Africa. *Journal of Cross-Cultural Psychology*, 44(3), 365–388. <https://doi.org/10.1177/0022022112443856>
- van der Linden, D., Dunkel, C. S., Figueredo, A. J., Gurven, M., von Rueden, C., & Woodley of Menie, M. A. (2018). How universal is the general factor of personality? An analysis of the Big Five in forager farmers of the Bolivian amazon. *Journal of Cross-Cultural Psychology*, 49(7), 1081–1097. <https://doi.org/10.1177/0022022118774925>
- Wasti, S. A., Lee, K., Ashton, M. C., & Somer, O. (2008). Six Turkish personality factors and the HEXACO model of personality structure. *Journal of Cross-Cultural Psychology*, 39(6), 665–684. <https://doi.org/10.1177/0022022108323783>
- Welzel, C., Brunkert, L., Kruse, S., & Inglehart, R. F. (2021). Non-invariance? An overstated problem with misconceived causes. *Sociological Methods & Research*. Advance online publication. <https://doi.org/10.1177/0049124121995521>
- Zeinoun, P., Daouk-öyry, L., Choueiri, L., & van de Vijver, F. J. R. (2017). A mixed-methods study of personality conceptions in the Levant: Jordan, Lebanon, Syria, and the West Bank. *Journal of Personality and Social Psychology*, 113(3), 453–465. <https://doi.org/10.1037/pspp0000148>
- Zhang, R., & Li, L. M. W. (2014). The acculturation of relational mobility: An investigation of Asian Canadians. *Journal of Cross-Cultural Psychology*, 45(9), 1390–1410. <https://doi.org/10.1177/0022022114542850>